



KALDRIVON

# Kaldrivon MiniSMO Quick Start

Version 0.3.1 · Windows 11 x64 · © 2026 Kaldrivon

This guide gets you from download to reading real NETCONF data in about fifteen minutes. The full User Guide and the in-app Help Center (press **F1**) cover everything in more detail.

## 1. Install

1. Download `Kaldrivon-MiniSMO-0.3.1-Setup.exe` and its `.sha256` file.
2. Optional: check the hash in PowerShell with `Get-FileHash .\Kaldrivon-MiniSMO-0.3.1-Setup.exe` and compare it with the `.sha256` file.
3. Run the installer, accept the defaults ( `C:\Program Files\Kaldrivon\MiniSMO` ) and choose whether to add a desktop shortcut.
4. Select **Finish** with *Launch Kaldrivon MiniSMO* ticked.

You do not need Python, .NET or anything else. Requirements: Windows 11 x64, 4 GB RAM (8 GB recommended).

## 2. First launch

MiniSMO starts its local backend automatically (no console window) and opens the **Dashboard**. On a new installation you see:

- `Community | Managed NFs 0/5` under the page title and in the title bar
- **ADD EXTERNAL NF** (the Demo Network is created under *SIMULATION > Demo Network*)

If MiniSMO reports that the backend could not start, select **Open startup troubleshooting**.

Screenshot placeholder: **Dashboard on first launch: Community, Managed NFs 0/5**  
To be captured from the Windows 0.3.1 build (docs/images/screenshots/dashboard-first-run.png)  
before public release.

## 3. Create the Demo Network

Open *SIMULATION > Demo Network* and select **CREATE DEMO NETWORK**.  
MiniSMO creates 14 simulated NFs: O-CU-CP-001, O-CU-UP-001, O-DU-001, O-DU-

002 and O-RU-001 to O-RU-010. O-DU-001 serves O-RU-001 to 005, O-DU-002 serves O-RU-006 to 010.

Simulated NFs do not count against your license: the indicator still shows **Managed NFs 0/5**.

Things to try:

- *NETWORK > Topology & Inventory* to see the three layers.
- *SIMULATION > Scenarios*, run **CELL OUTAGE**, then look at *FCAPS > Alarms* and the topology. Select **STOP** when done.
- *FCAPS > Performance* to watch PRB, throughput and CPU move.

## 4. Add an external NF

You need a device (or NETCONF test server) that accepts NETCONF over SSH, usually on TCP port 830, and a username and password or SSH key for it.

1. Select **ADD EXTERNAL NF** on the Dashboard or on *NETWORK > Network Functions*.
2. Enter a **Name** (for example `LAB-O-RU-01`), **NF Type**, **Hostname/IP**, **NETCONF Port** (830), **Authentication**, **Username** and **Password**.
3. Select **TEST CONNECTION** (next section).
4. Select **SAVE**. The indicator shows **Managed NFs 1/5**. MiniSMO then integrates the NF: it connects, starts FM, CM and PM monitoring and points the NF's VES destination at the collector. A dialog shows each step. If the device is unreachable, the NF is saved and shows **DISCONNECTED**.

Passwords are encrypted with Windows DPAPI and never shown or logged. Automatic integration can be turned off in *SYSTEM > Settings > General*, section **Integration**.

## 5. Test the connection

**TEST CONNECTION** checks six stages in order and marks each one ✓ or ✗:

| Stage             | If it fails, check                                                 |
|-------------------|--------------------------------------------------------------------|
| Host Resolution   | Spelling, DNS, VPN; try the IP address                             |
| TCP Connection    | Device reachable, port 830 open, firewalls                         |
| SSH               | The port really is SSH                                             |
| Authentication    | Username and password or key; the sign-in method the device allows |
| NETCONF subsystem | NETCONF enabled on the device for that port                        |

| Stage        | If it fails, check                                        |
|--------------|-----------------------------------------------------------|
| Server Hello | Device NETCONF service healthy; raise the connect timeout |

A full pass shows the NETCONF version (1.0 or 1.1) and the number of capabilities. **CONNECTED** is only shown after a real NETCONF hello exchange.

## 6. Run GET

1. Open *FCAPS > Configuration* and select your NF.
2. Open the **NETCONF read** tab and select **GET**.

The result shows status, message ID, session ID, timestamp, duration and the full request and reply XML (as a tree or as raw XML). The **Capabilities** tab of the NF in *NETWORK > Network Functions* lists every capability and YANG module the device advertised.

## 7. Run GET-CONFIG

On the **NETCONF read** tab, pick a **Datastore** in the toolbar and select **GET-CONFIG**. Only datastores the device advertises can be selected (*running* always; *candidate* and *startup* only if supported).

Every NETCONF message is recorded in *LOG > NETCONF Log*, where you can search, filter, copy and export it.

Configuration changes on an external NF are off until you choose **ENABLE CHANGES...** for that NF on *FCAPS > Configuration*. Each change is reviewed and confirmed. Simulated NFs can be edited directly.

## 8. Monitoring

Open the NF's **Monitoring** tab in *NETWORK > Network Functions* to see FM, CM and PM state, the methods in use, VES events and file-based PM. Alarms appear in *FCAPS > Alarms*, PM in *FCAPS > Performance*.

## 9. License limit

Kaldrivon Mini SMO Community manages **5 external NFs**. Connected and disconnected NFs both count; simulated NFs never count; deleting an external NF frees its slot. When you reach the limit, **ADD EXTERNAL NF** reports *Community supports up to 5 external managed NFs (5/5 in use)*.

Community also includes a small daily allowance of each Pro feature (*TOOLS* pages, marked **PRO**). *SYSTEM > License* lists it with today's use.

For more capacity and the Pro features, install a Kaldrivon `.kllicense` file on *SYSTEM > License* with **INSTALL LICENSE**. If a license expires, nothing is deleted; capacity returns to 5.

## 10. Where to find Help

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- **F1** on any page, or the **?** button in a page header, opens the matching help topic.
- **Help** at the bottom of the navigation pane opens the Help Center with search. Try *port 830*, *hello* or *license*.
- The **User Guide** (HTML and PDF) is installed with the application: **About > OPEN USER GUIDE**.
- Help works without an Internet connection.
- Support: [info@kaldrivon.com](mailto:info@kaldrivon.com) · <https://kaldrivon.com>